

A DEVELOPMENT OF PREDICTING STUDENTS ACADEMIC PERFORMANCE USING DECISION TREE ALGORITHM

By

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Introduction

__ _ Traditional student monitoring is reactive, which leads to late intervention and high dropout rates. Our solution uses Decision Tree Machine Learning to continuously approximate the risk to students by analyzing grade, appearance, and engagement data with 85%+ accuracy. It improves timely support, retention, and changes education through data-operated insights.

Statement of the Problem

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Reactive monitoring → late intervention

High dropout rate and bad results

The manual system lacks predictive power

Aim and Objectives

— — — • **Aim:**

To Design a predictive model using the Decision Tree algorithm to forecast students' academic performance, facilitating early identification of at-risk students and enabling data-driven interventions

• **Objectives:**

1. To collect and preprocess student academic data.
2. To design an accurate Decision Tree prediction model
3. To evaluate the model's accuracy and effectiveness.
4. To identify key performance predictors.

Literature Review

Decision Trees are proven effective (85-90% accuracy) for academic prediction (Romero et al., 2020).

Key predictors: Attendance, grades, LMS engagement (Kumar et al., 2022).

Advantage: Interpretable rules > "black-box" models (e.g., neural networks).

Gaps: Need for non-academic factors (mental health, socioeconomics).

Literature Review

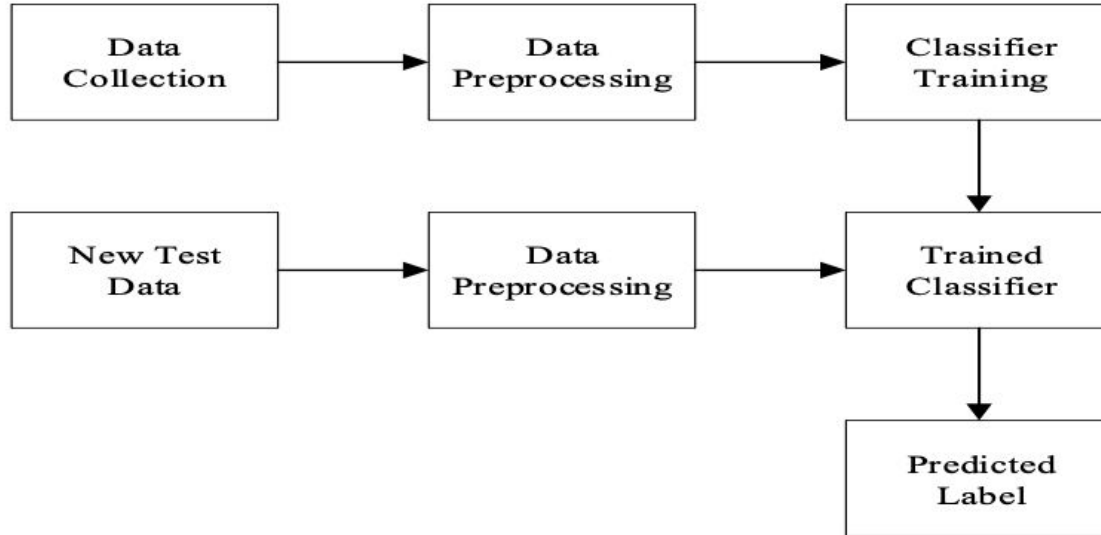
Authors (Year)	Strengths	Gaps / Limitations
Aluja-Banet et al. (2023) - Predicting academic performance in European universities using C4.5 Decision Trees	Achieved 87.3% accuracy; identified attendance as a critical predictor.	Limited to European contexts; did not address model bias across demographics.
Kumar et al., (2022) - Decision Trees for online learning dropout prediction	High AUC (91%) in MOOC data; emphasized forum participation.	Ignored non-digital factors (e.g., mental health).
Chen & Gao (2022) - Random Forest model with IoT data (dorm electricity, cafeteria spending)	89.1% accuracy; incorporated behavioral data.	Privacy concerns with IoT; lacked longitudinal validation.
Martinez et al. (2021) - Early dropout prediction in K-12 using boosted Decision Trees	Predicted dropout as early as 6th grade (76% precision)	Limited to Texas schools; did not account for curriculum changes.
Wang et al. (2023) - LSTM vs. Decision Trees for temporal academic data	Neural networks achieved 92.1% accuracy.	Black-box model; required large datasets (>10,000 records)

Methodology

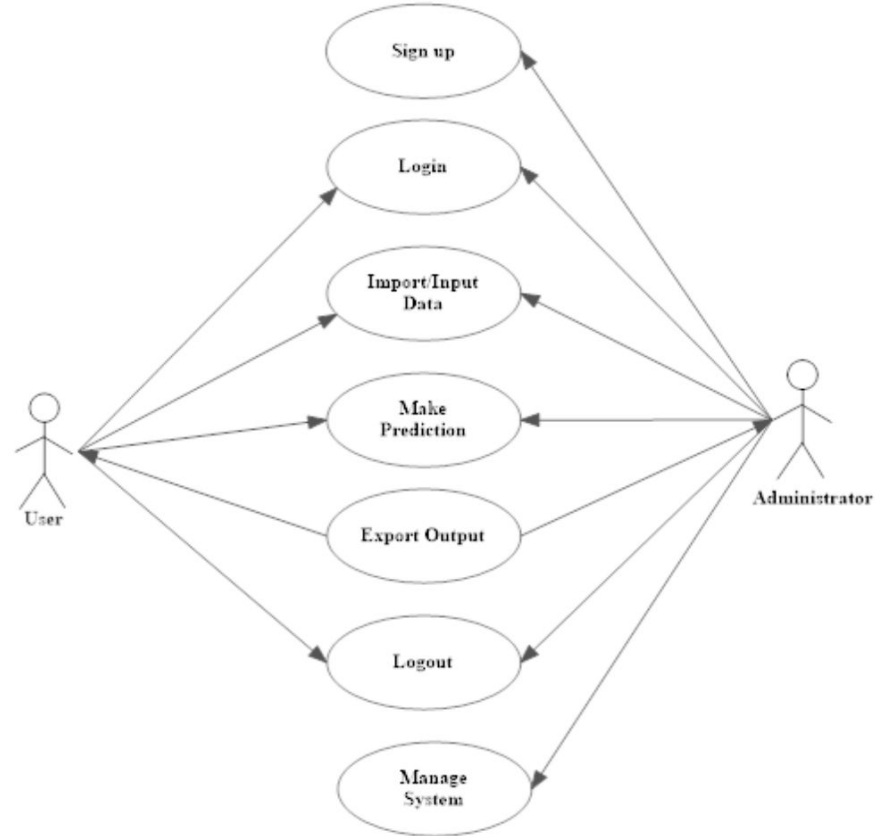
This study combines quantitative data analysis with Agile development principles to create a Decision Tree-based prediction system. The methodology follows five iterative phases: requirements analysis, data preprocessing, model development, evaluation, and deployment. This approach ensures flexibility, continuous improvement, and alignment with educators' needs throughout the project lifecycle.

Methodology (DATA FLOW DIAGRAM)

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Methodology (USE CASE UML DIAGRAM)



Conclusion

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This study demonstrates how Decision Trees can transform student performance prediction through data-driven insights (85%+ accuracy). By analyzing grades, attendance, and engagement data, the model enables early intervention while maintaining transparency for educators.

References

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